GOOGLE SHEETS ANALYSIS DOCUMENTATION

Seoul Bike Rental Demand Analysis | Data Preparation, Pivot Tables & KPIs

Purpose

This document describes the Google Sheets workflow used in the Seoul Bike Rental Demand Analysis project. It documents the derived fields, lookup logic, Pivot Tables, and KPI summaries used to explore hourly, seasonal, and weekday-versus-weekend rental patterns.

Workbook Structure

Sheet	Purpose
Raw_Data	Stores the imported dataset and preserves the original columns.
Analysis	Contains the working copy of the data and derived analytical fields.
Hour_Lookup	Maps hour ranges to descriptive time-of-day categories.
Pivot_Analysis	Contains the three Pivot Tables used for exploratory analysis.
Executive_Summary	Summarizes KPI values and key findings for reporting.

1. Data Preparation & Derived Fields

The Analysis sheet was used as the working table. The following derived fields were added to support comparisons by weekday/weekend, time of day, and operating status.

Day of Week

Purpose: derive the weekday name from the date field in column A.

```
=TEXT(A2,"dddd")
```

Day Type

Purpose: classify each date as Weekday or Weekend. WEEKDAY(date,2) numbers Monday as 1 through Sunday as 7.

```
=IF(WEEKDAY(A2,2)<=5,"Weekday","Weekend")
```

Hour Category

Purpose: group numeric hours into descriptive time-of-day categories using the Hour_Lookup sheet. The lookup table uses approximate matching, so the first column must be sorted in ascending order.

```
=VLOOKUP(C2,Hour_Lookup!$A$2:$B$6,2,TRUE)
```

Start Hour	Hour Category
0	Overnight
6	Morning
9	Daytime
16	Evening Peak
20	Late Evening

Operational Status

Purpose: convert the Functioning Day field into an operating-status label.

```
=IF(N2="Yes","Operating","Not Operating")
```

These formulas were filled down through the 8,760 hourly observations (rows 2–8761), as described during project work.

2. Pivot Table Analysis

Three Pivot Tables were created on the Pivot_Analysis sheet. The value field for each table is the average of Rented Bike Count.

Pivot 1 — Average Rentals by Hour

Rows: Hour. Values: AVERAGE of Rented Bike Count. Filter: Operational Status = Operating. This table identifies hourly demand peaks and lower-demand periods.

Observed results: 6:00 PM is the peak hour at approximately 1,554.02 average rentals; 8:00 AM is a secondary peak at 1,050.23; 4:00 AM is the lowest hour at 137.49.

Pivot 2 — Average Rentals by Hour and Season

Rows: Hour. Columns: Seasons. Values: AVERAGE of Rented Bike Count. Filter: Operational Status = Operating. This table compares hourly patterns and overall average rentals across seasons.

Season	Overall average rentals
Summer	1,034.07
Autumn	924.11
Spring	746.25
Winter	225.54

At 6:00 PM, seasonal average rentals were 2,135.14 in Summer, 1,983.33 in Autumn, 1,689.31 in Spring, and 438.30 in Winter.

Pivot 3 — Average Rentals by Hour and Day Type

Rows: Hour. Columns: Day Type (Weekday, Weekend). Values: AVERAGE of Rented Bike Count. Filter: Operational Status = Operating. This table compares time-of-day demand between weekdays and weekends.

Hour	Weekday	Weekend
8:00 AM	1,306.02	420.79
12:00 PM	700.09	780.13
3:00 PM	790.23	1,022.60
6:00 PM	1,724.05	1,135.60

3. KPI Summary

KPI	Value shown	Definition / interpretation
Total Rentals	6,172,314	Sum of Rented Bike Count in the dashboard.
Average Rentals per Operating Hour	729.16	Average Rented Bike Count for observations classified as Operating.
Peak Hour	18:00 (1,554.02)	Hour with the highest average rental count in Pivot 1.

The Total Rentals KPI is a sum, while the other values are averages or a peak-hour summary. These measures should not be interpreted as equivalent metrics.

4. Workflow Summary

• Import and preserve the original dataset in Raw_Data.

• Create derived fields in Analysis and fill formulas down the dataset.

• Use Hour_Lookup for approximate VLOOKUP categorization.

- Build Pivot Tables in Pivot_Analysis to summarize demand patterns.
- Summarize key KPI values and findings in Executive_Summary.
- Use Tableau Public to present the key metrics and visual comparisons in a dashboard.

Documentation note: the field formulas and Pivot Table configurations reflect the workflow described for this project. Verify formula column references against the current workbook if the source columns have been rearranged.